



Palo Alto Networks Cybersecurity Academy –Setting up a PA220 Firewall for a SOHO Environment

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Purpose

The purpose of this lab is to set up a PA220 for a small-scale SOHO network, which is the primary use case for the device. The lab covers skills such as setting up DHCP services and trust boundaries on a router, which are essential for basic network functions and security.

Background

SOHO, which stands for Small Office/Home Office, is a network type commonly used by individuals or small businesses with less than 10 employees. This network type commonly uses smaller-scale routers, switches, and firewalls compared to their large enterprise counterparts. SOHO networks provide numerous advantages to teams of 1-10 people as they are easier to set up and are more affordable than full-size network equipment. SOHO networks often only have a single router, and may contain switches, wireless access points, and end devices such as computers and printers.

The Palo Alto Networks PA-220 next-generation firewall, contrary to their core product, is designed for small organizations or branch offices and includes the following main features: active/passive and active/active high availability (HA), passive cooling (no fans) to reduce noise and power consumption, eight Ethernet ports, and dual power adapters for power redundancy. The PA-220 firewall enables you to secure your organization through advanced visibility and control of applications, users, and content. Marketed as a NGFW, Next-generation firewall. The PA220 uses machine learning to identify attacks instead of relying on simple signature check like traditional firewalls. This technology allows the PA220 to identify undocumented threats and brand-new exploits without intervention from Palo Alto networks themselves. The PA220 also prevents threats by filtering URLs and securing against DNS-based attacks.

Since the PA220 is a hardware firewall, meaning it's a physical device, it is a more tangible device compared to a virtual software firewall. By using this legacy form factor, the PA220 can provide a higher degree of reliability as the physical device can be troubleshooted by the end user. With a software firewall, security is managed remotely by a 3rd party company, meaning that any issues with the outside company could leave software firewalls users vulnerable for hours or even days. Hardware firewalls put this control in the hands of the users, allowing security to be implemented immediately by an experienced technician.

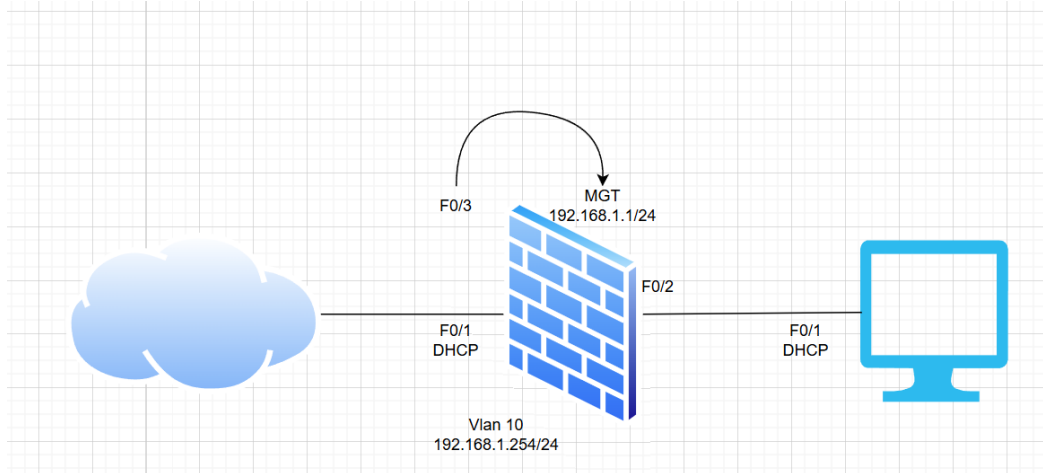
Palo Alto firewalls run on an operating system called PAN-OS. PAN-OS can be controlled through two methods: a Graphical User Interface (GUI) and a Command-Line Interface (CLI). The GUI is accessible through an HTTP connection and displays in any modern web browser. The HTTP connection is available through the firewall's MGT port and by default, is accessible at <https://192.168.1.1>. The firewall has a default username and password of admin.

Lab Summary

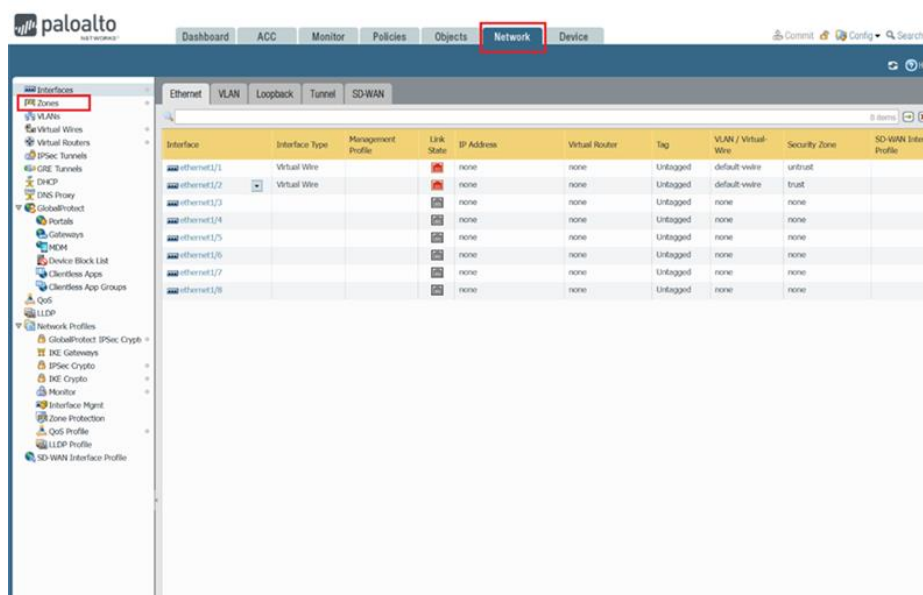
In this lab, we configured our firewall to connect to a DHCP-enabled ISP router

and configured this traffic to be untrusted by default. We then configured the remainder of the router ports to be connected by a single VLAN, have trusted traffic, and be DHCP clients served by the firewall

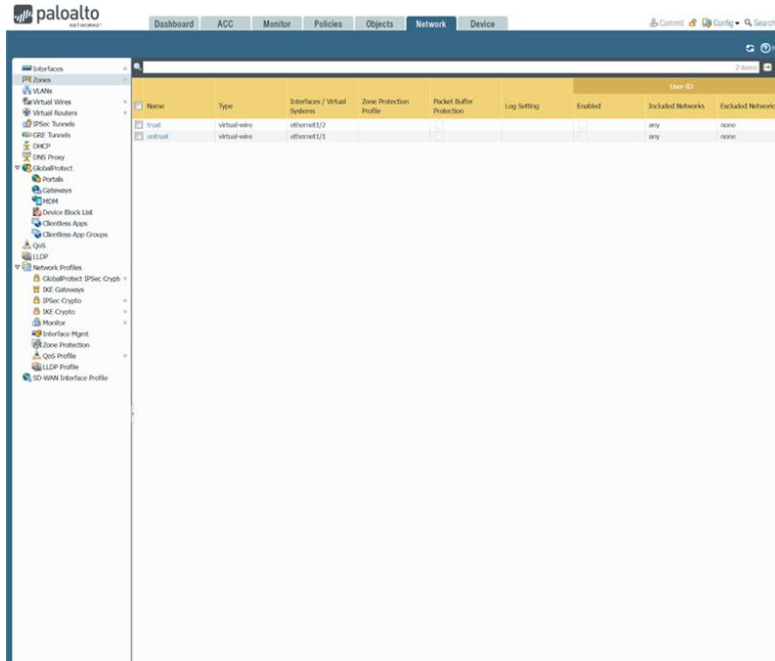
Network Diagrams



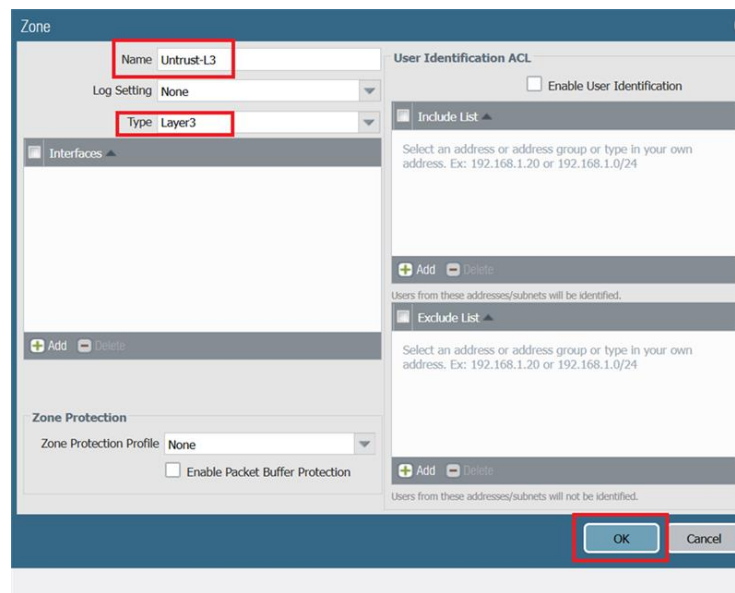
Lab Commands



1. When in your Web GUI go to the Network tab as highlighted in the image above. Then click the Zone section on the left hand side as highlighted in the image above.



1. Then once on a screen like this click the add button on the bottom of the page.



1. Change the Name to Untrust-L3 as highlighted in the image above. Also modify the type to Layer3. Then, Click the OK button as highlighted in the image above.

							3 items		
☐	Name	Type	Interfaces / Virtual Systems	Zone Protection Profile	Packet Buffer Protection	Log Setting	User-ID		
							Enabled	Included Networks	Excluded Networks
☐	trust	virtual-wire	ethernet1/2		☐		☐	any	none
☐	untrust	virtual-wire	ethernet1/1		☐		☐	any	none
☒	Untrust-L3	layer3			☐		☐	any	none

4. Then click the same add button to create another zone.

Zone

Name Trust-L3

Log Setting None

Type Layer3

Interfaces

Add

Delete

Zone Protection

Zone Protection Profile None

Enable Packet Buffer Protection

User Identification ACL

Enable User Identification

Include List

Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24

Add

Delete

Exclude List

Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24

Add

Delete

OK

Cancel

5. Change the Name to Trust-L3 as highlighted in the image above. Also modify the type to Layer3. Then, Click the OK button as highlighted in the image above.

4 items								
Name	Type	Interfaces / Virtual Systems	Zone Protection Profile	Packet Buffer Protection	Log Setting	User ID		
Enabled	Included Networks	Excluded Networks						
☐	trust	virtual-wire	ethernet1/2	☐		☐	any	none
☐	untrust	virtual-wire	ethernet1/1	☐		☐	any	none
☐	Untrust-L3	layer3		☐		☐	any	none
☒	Trust-L3	layer3		☐		☐	any	none

5. Then click the same add zone button to create another zone

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Zone

Name **Trust-L2**

Log Setting None

Type **Layer2**

Interfaces

+ Add - Delete

Zone Protection

Zone Protection Profile None

☐ Enable Packet Buffer Protection

User Identification ACL

☐ Enable User Identification

Include List

Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24

+ Add - Delete

Users from these addresses/subnets will be identified.

Exclude List

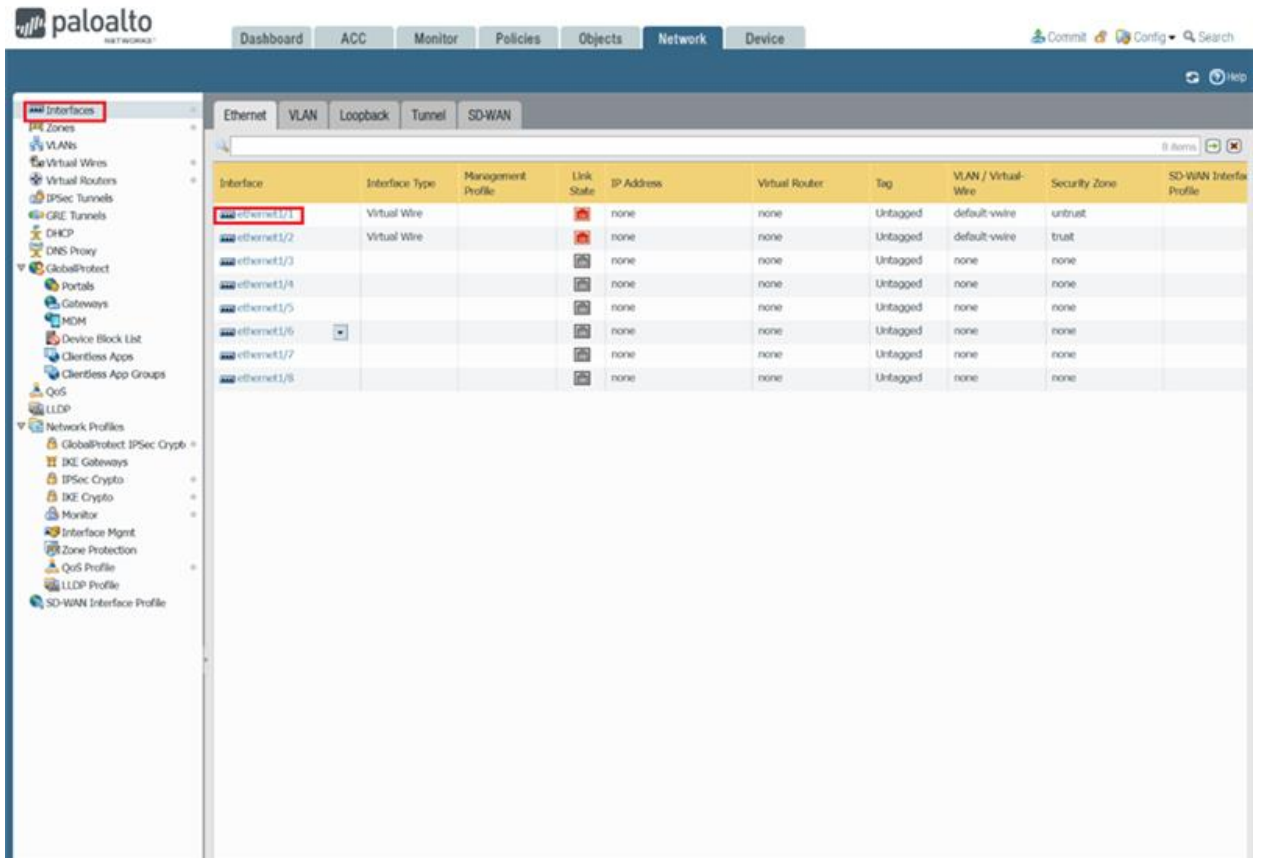
Select an address or address group or type in your own address. Ex: 192.168.1.20 or 192.168.1.0/24

+ Add - Delete

Users from these addresses/subnets will not be identified.

OK Cancel

5. Change the Name to Trust-L2 as highlighted in the image above. Also modify the type to Layer2. Then, Click the OK button as highlighted in the image above.



- Click the Interfaces tab on the left hand side as highlighted in the image above. Then select the Ethernet1/1 option as highlighted in the image above.

The screenshot shows the 'Ethernet Interface' configuration window for 'ethernet1/1'. The 'Interface Type' is set to 'Layer3', 'Netflow Profile' is 'None', 'Virtual Router' is 'default', and 'Security Zone' is 'Untrust-L3'.

Interface Name: ethernet1/1

Comment:

Interface Type: Layer3

Netflow Profile: None

Config: IPv4 | IPv6 | SD-WAN | Advanced

Assign Interface To:

Virtual Router: default

Security Zone: Untrust-L3

OK Cancel

5. Configure the settings as highlighted in the image above.

The screenshot shows the 'Ethernet Interface' configuration window for 'ethernet1/1'. The 'IPv4' tab is selected. The following settings are highlighted with red boxes:

- Interface Name:** ethernet1/1
- Interface Type:** Layer3
- Netflow Profile:** None
- Config Tab:** IPv4
- Enable SD-WAN:** ☐
- Type:** ☐ Static ☐ PPPoE ☒ **DHCP Client**
- Enable:** ☒
- Automatically create default route pointing to default gateway provided by server:** ☒
- Send Hostname:** ☐ system-hostname
- Default Route Metric:** 10
- Buttons:** OK, Cancel

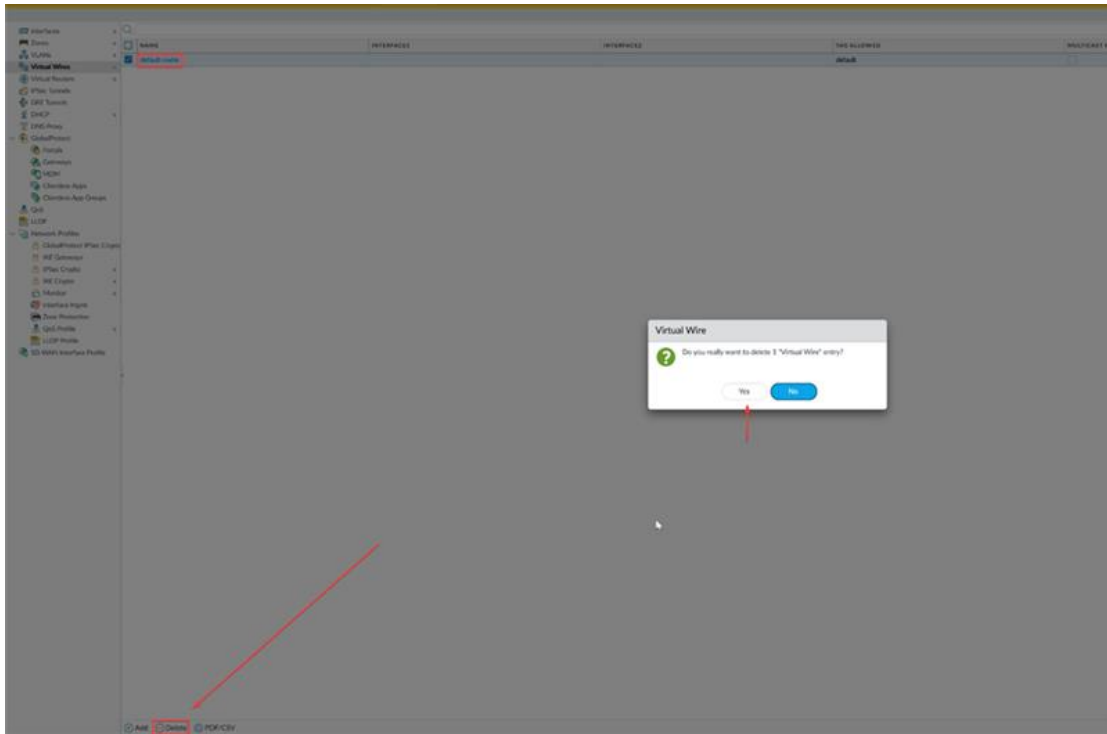
Below the configuration fields, there is a link: [Show DHCP Client Runtime Info](#).

5. Click the IPV4 tab and modify the settings as highlighted in the image above. Then press the OK button as highlighted in the image above.

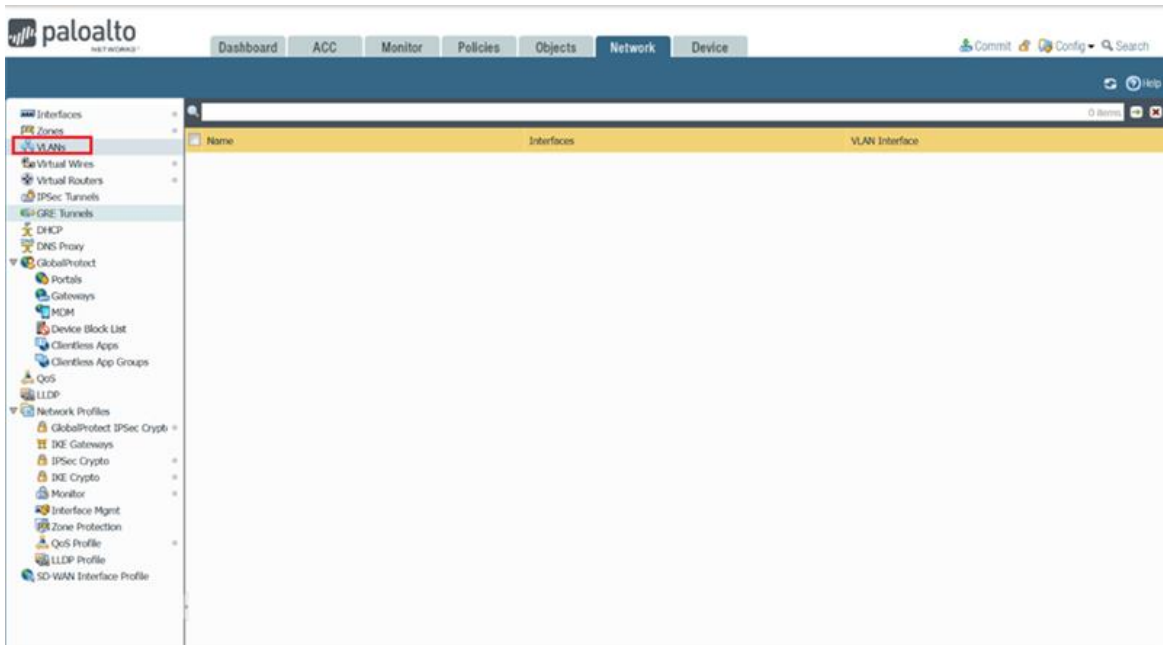
The screenshot shows the 'Interfaces' configuration page in the Palo Alto Networks GUI. The 'Virtual Routers' section is highlighted in the left sidebar. The main table displays the configuration for the 'default' interface.

Name	Interfaces	Configuration	RIP	OSPF	OSPFv3	BGP	Multicast	Runtime Stats
default	ethernet1/1	ECMP status: Disabled						More Runtime Stats

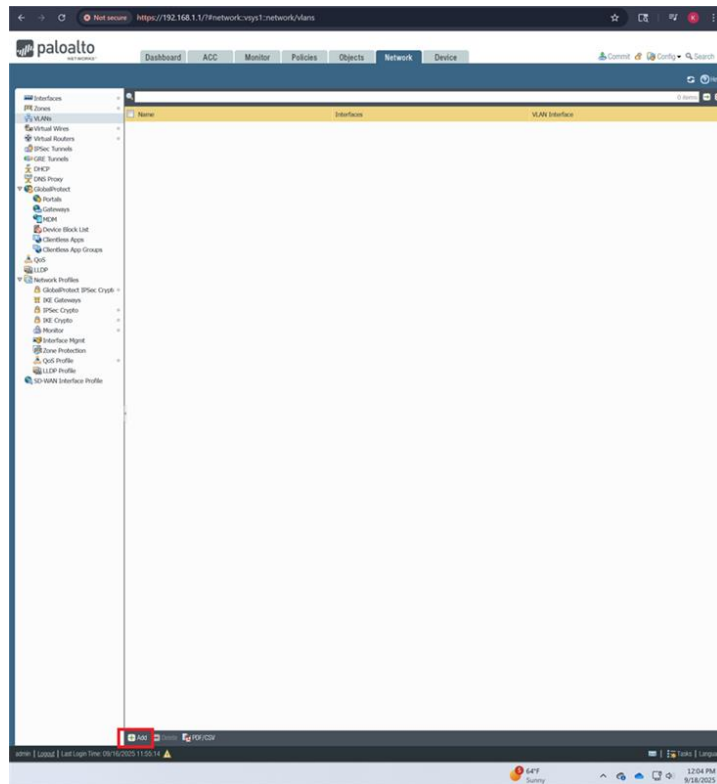
- Click the virtual interface tab on the left side as highlighted in the image above.



- Click on the delete button on the bottom of the GUI, select “default”, and then confirm the delete selection.



- Click the VLANs tab on the left as highlighted in the image above.

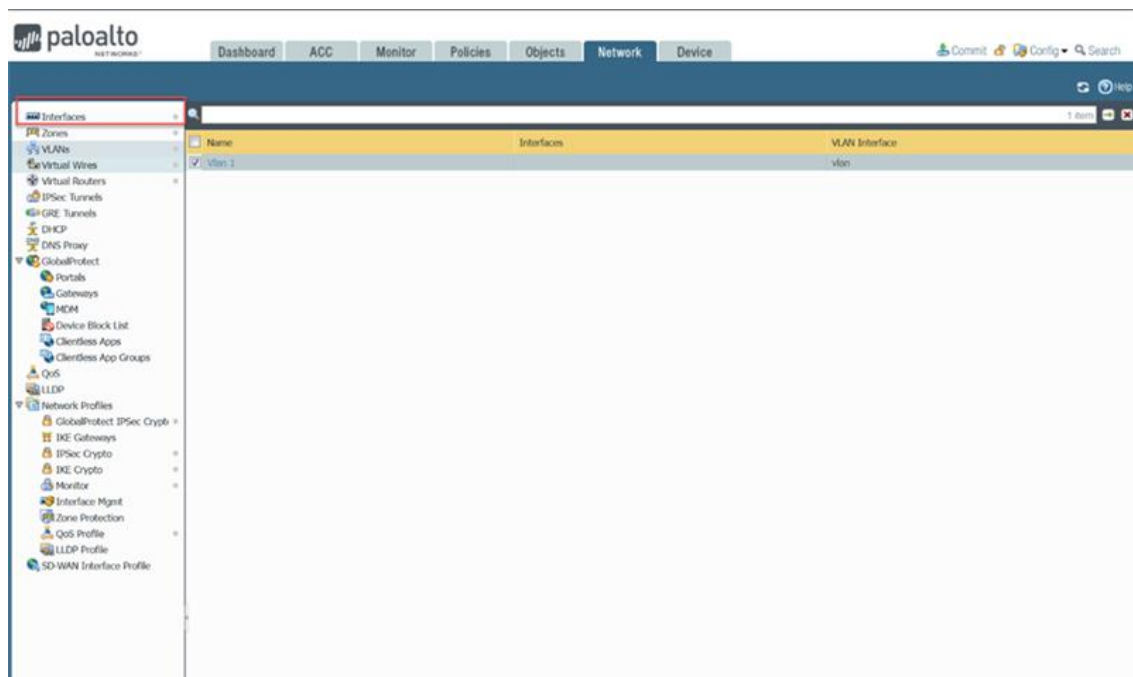


- Click the add button on the bottom of the GUI as highlighted in the image above.

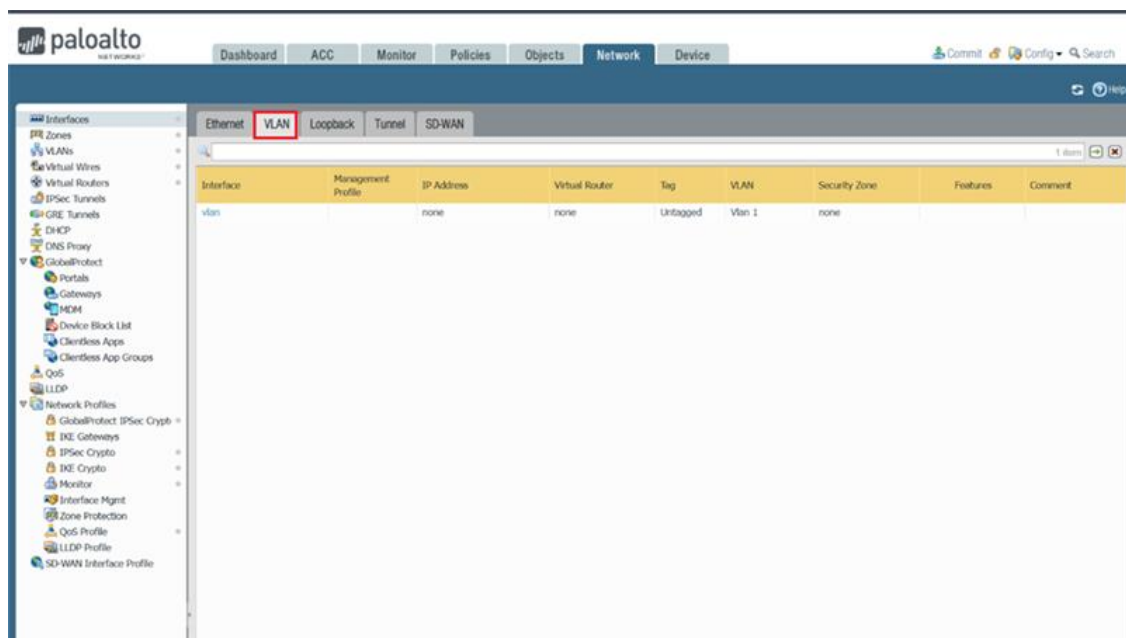
The screenshot shows the 'VLAN' configuration dialog box. The 'Name' field is set to 'Vlan 1' and the 'VLAN Interface' dropdown is set to 'vlan'. Below these fields is a section titled 'Static MAC Configuration' which contains a table with columns 'Interfaces', 'MAC Address', and 'Interface'. At the bottom of the dialog, there are 'Add' and 'Delete' buttons for both the table and the overall configuration. The 'Add' button at the bottom right is highlighted with a red box.

Interfaces	MAC Address	Interface
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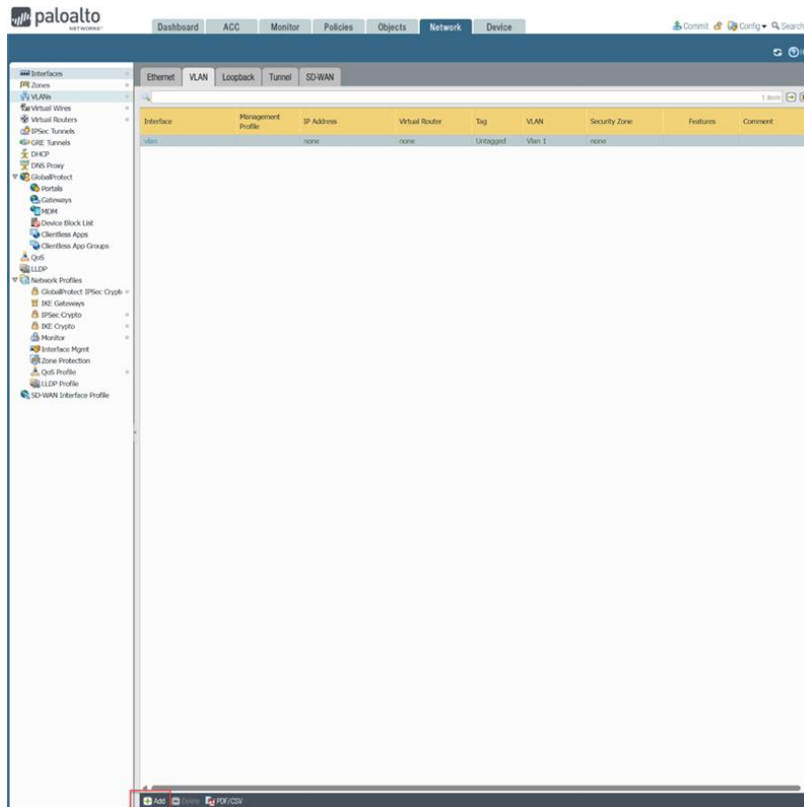
5. Modify the VLAN with the settings as highlighted in the image above. Then, select the OK button.



5. Click the Interfaces tab on the left as highlighted in the image above.



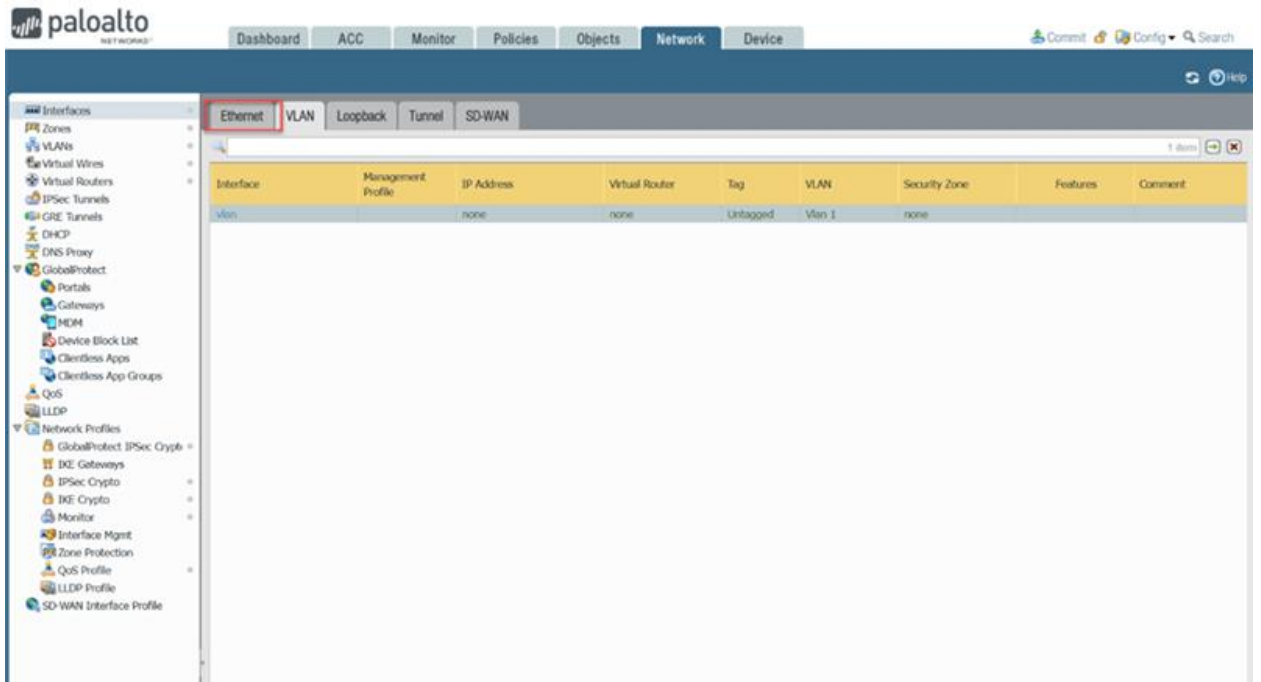
5. Click the VLAN tab as highlighted in the image above.



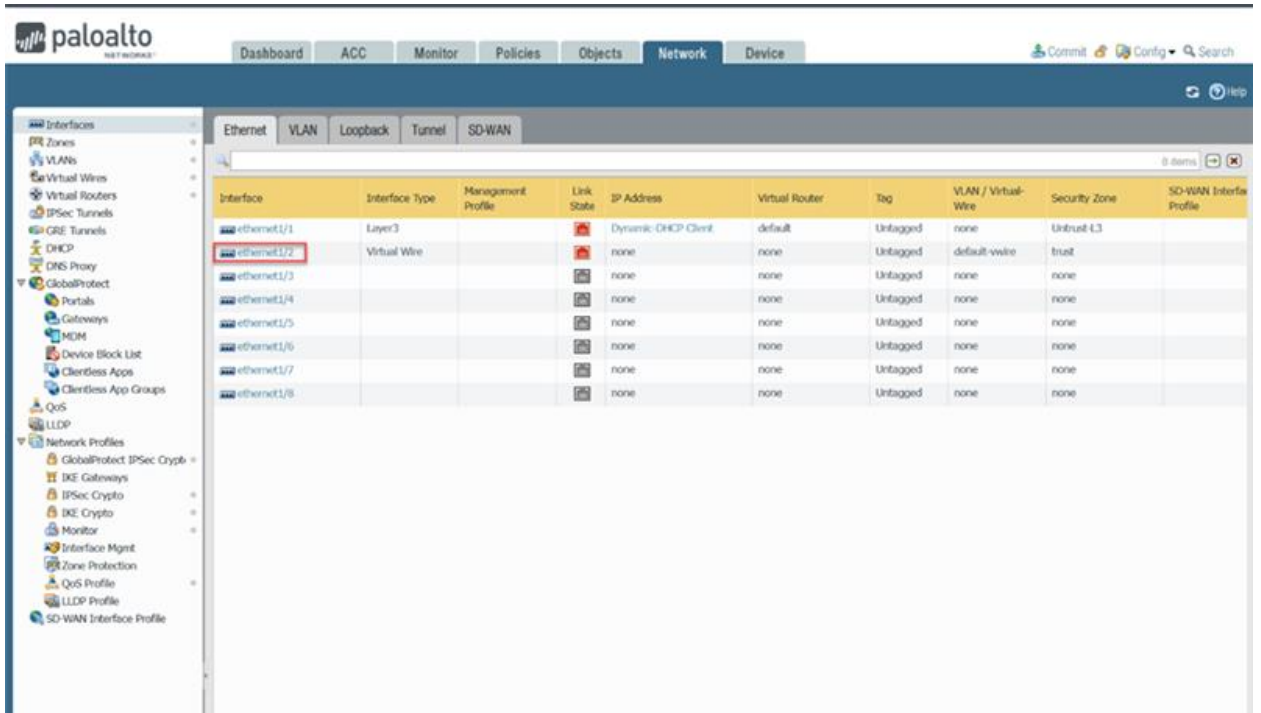
5. Click the add button as highlighted in the image above.

The screenshot shows the 'VLAN Interface' configuration window. The 'Interface Name' field is set to 'vlan' and the 'VLAN' dropdown is set to 'Vlan 1'. The 'OK' button is highlighted with a red box.

5. Modify the settings as highlighted in the image above.



5. Select the Ethernet tab as highlighted in the image above.



5. Click the Ethernet1/2 as highlighted in the image above.

Ethernet Interface

Interface Name: ethernet1/2

Comment:

Interface Type: **Layer2**

Netflow Profile: **None**

Config **Advanced**

Assign Interface To

VLAN: **Vlan 1**

Security Zone: **Trust-L2**

OK **Cancel**

5. Modify the settings as highlighted in the image above. Then, select the OK button.

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Clientless Apps

Clientless App Groups

QoS

LLDP

Network Profiles

GlobalProtect IPSec Crypto

IKE Gateways

IPSec Crypto

IKE Crypto

Monitor

Interface Mgmt

Zone Protection

QoS Profile

LLDP Profile

SD-WAN Interface Profile

Ethernet

VLAN

Loopback

Tunnel

SD-WAN

8 items

Interface	Interface Type	Management Profile	Link State	IP Address	Virtual Router	Tag	VLAN / Virtual-Wire	Security Zone	SD-WAN Interface Profile
ethernet1/1	Layer3		Dynamic DHCP Client	default	Untagged	none	Trust-L3		
ethernet1/2	Layer2		none	none	Untagged	Vlan 1	Trust-L2		
ethernet1/3			none	none	Untagged	none	none		
ethernet1/4			none	none	Untagged	none	none		
ethernet1/5			none	none	Untagged	none	none		
ethernet1/6			none	none	Untagged	none	none		
ethernet1/7			none	none	Untagged	none	none		
ethernet1/8			none	none	Untagged	none	none		

5. Select the Ethernet1/3 as highlighted in the image above.

Ethernet Interface

Interface Name: **ethernet1/3**

Comment:

Interface Type: **Layer2**

Netflow Profile: **None**

Config **Advanced**

Assign Interface To

VLAN: **Vlan 1**

Security Zone: **Trust-L2**

OK **Cancel**

5. Modify the settings as highlighted in the image above.

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IKE Gateways

IPSec Crypto

IKE Crypto

Monitor

Interface Mgmt

Zone Protection

QoS Profile

LLDP Profile

SD-WAN Interface Profile

Ethernet

VLAN

Loopback

Tunnel

SD-WAN

8 items

Interface	Interface Type	Management Profile	Link State	IP Address	Virtual Router	Tag	VLAN / Virtual Wire	Security Zone	SD-WAN Interface Profile
ethernet1/1	Layer3			Dynamic DHCP Client	default	Untagged	none	Untrust-L3	
ethernet1/2	Layer2			none	none	Untagged	Vlan 1	Trust-L2	
ethernet1/3	Layer2			none	none	Untagged	Vlan 1	Trust-L2	
ethernet1/4				none	none	Untagged	none	none	
ethernet1/5				none	none	Untagged	none	none	
ethernet1/6				none	none	Untagged	none	none	
ethernet1/7				none	none	Untagged	none	none	
ethernet1/8				none	none	Untagged	none	none	

5. Select Ethernet1/4 as highlighted in the image above.

Ethernet Interface

Interface Name: **ethernet1/4**

Comment:

Interface Type: **Layer2**

Netflow Profile: **None**

Config Advanced

Assign Interface To

VLAN: **Vlan 1**

Security Zone: **Trust-L2**

OK Cancel

5. Modify the settings as highlighted in the image above.

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Dashboard ACC Monitor Policies Objects **Network** Device

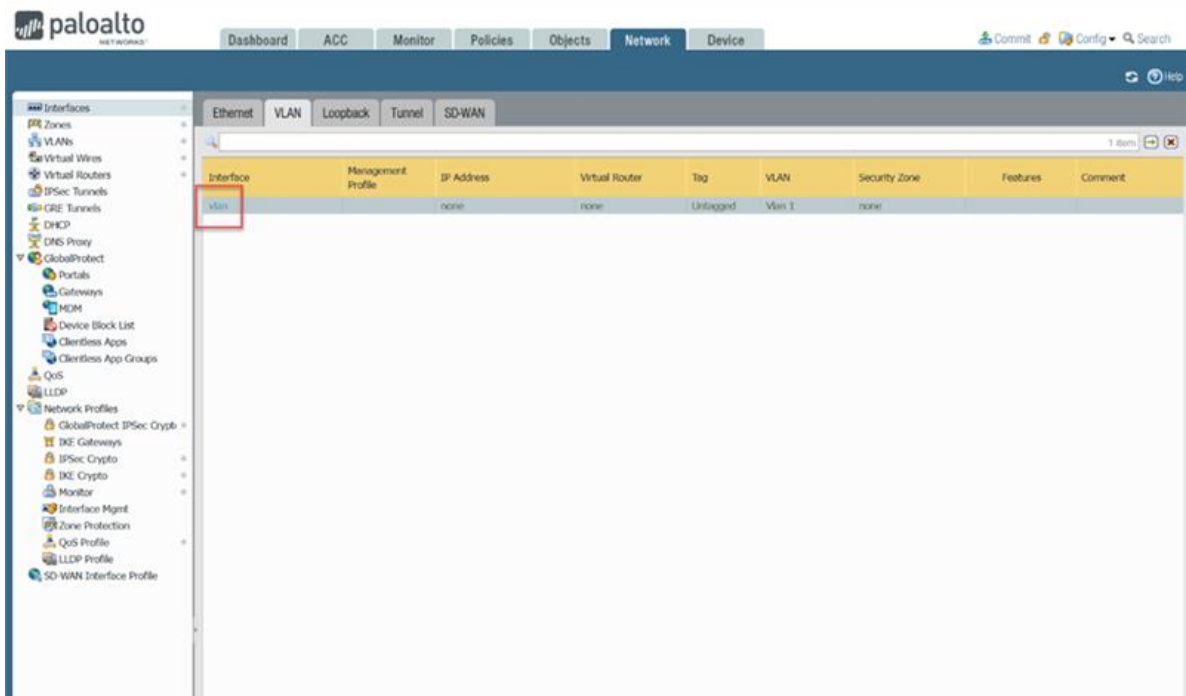
Commit Config Search

Interfaces VLANs Virtual Wires Virtual Routers IPsec Tunnels GRE Tunnels DHCP DNS Proxy GlobalProtect Portals Gateways MDM Device Block List Clientless Apps Clientless App Groups QoS LLDP Network Profiles GlobalProtect IPsec Crypto IKE Gateways IPsec Crypto IKE Crypto Monitor Interface Mgmt Zone Protection QoS Profile LLDP Profile SD-WAN Interface Profile

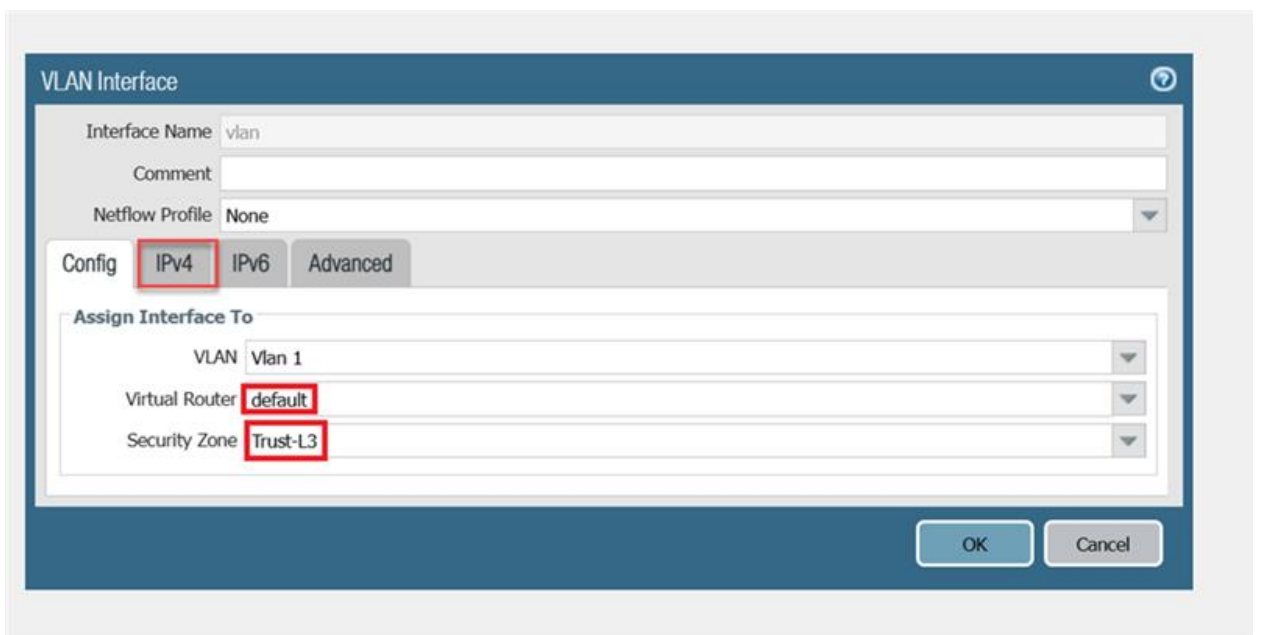
Ethernet **VLAN** Loopback Tunnel SD-WAN

Interface	Interface Type	Management Profile	Link State	IP Address	Virtual Router	Tag	VLAN / Virtual-Wire	Security Zone	SD-WAN Interface Profile
ethernet1/1	Layer3		Dynamic DHCP Client	default	Untagged	none	Trust-L3		
ethernet1/2	Layer2		none	none	Untagged	Vlan 1	Trust-L2		
ethernet1/3	Layer2		none	none	Untagged	Vlan 1	Trust-L2		
ethernet1/4	Layer2		none	none	Untagged	Vlan 1	Trust-L2		
ethernet1/5			none	none	Untagged	none	none		
ethernet1/6			none	none	Untagged	none	none		
ethernet1/7			none	none	Untagged	none	none		
ethernet1/8			none	none	Untagged	none	none		

5. Select the VLAN tab as highlighted in the image above.



5. Select "vlan" not "vlan.10" as highlighted in the image above.



5. Modify the settings as highlighted in the image above. Then Select the IPv4 tab as highlighted in the image above.

VLAN Interface

Interface Name: vlan

Comment:

Netflow Profile: None

Config | IPv4 | IPv6 | Advanced

Type: ☒ Static ☐ DHCP Client

IP

+ Add - Delete ↻ Move Up ↻ Move Down

OK Cancel

- Click the DHCP Client option as highlighted in the image above.

VLAN Interface

Interface Name: vlan

Comment:

Netflow Profile: None

Config | IPv4 | IPv6 | Advanced

Type: ☐ Static ☒ DHCP Client

☒ Enable

☒ Automatically create default route pointing to default gateway provided by server

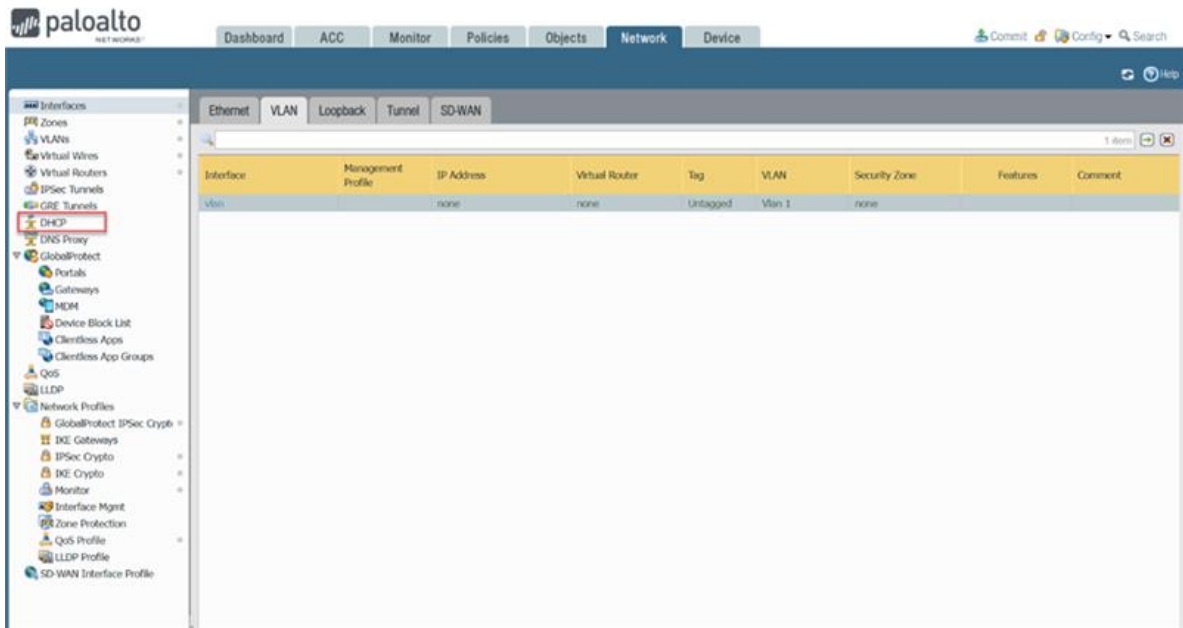
☐ Send Hostname: system-hostname

Default Route Metric: 10

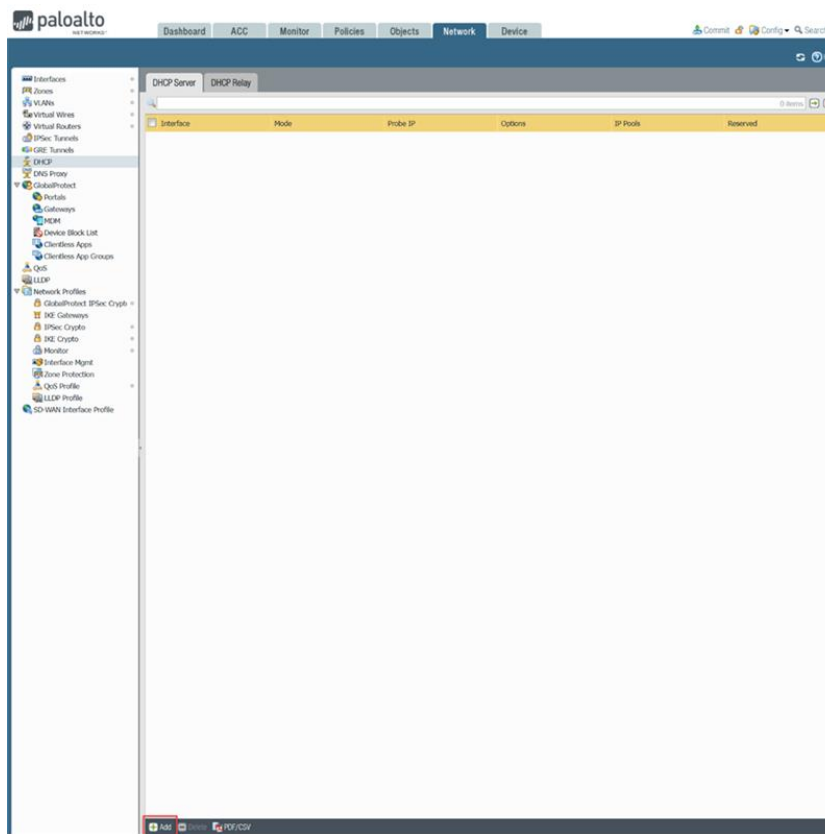
Show DHCP Client Runtime Info

OK Cancel

- Modify the settings as highlighted in the image above. Then select the OK button as highlighted in the image above.



5. Select the DHCP section on the left side of the GUI as highlighted in the image above.



5. Select the Add button on the bottom of the GUI as highlighted in the image above.

Interface: vlan

Mode: enabled

Lease Options

☐ Ping IP when allocating new IP

Lease: ☒ Unlimited ☐ Timeout

IP Pools	Reserved Address	MAC Address
192.168.1.20, 192.168.1.0/24 or 192.168.1.10-192.168.1.20	192.168.1.20	xx:xx:xx:xx:xx:xx (Optional MAC Address)

Buttons: Add, Delete, OK, Cancel

5. Modify the settings as highlighted in the image above. Then, select the Options tab as highlighted in the image above.

Interface: vlan

Mode: enabled

Lease Options

☐ Ping IP when allocating new IP

Lease: ☒ Unlimited ☐ Timeout

Inheritance Source: ethernet1/1

Check inheritance source status

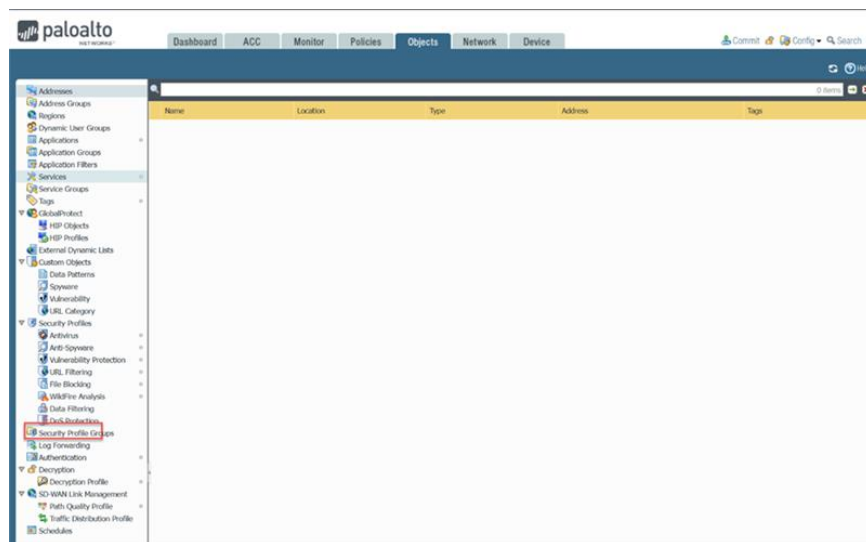
Option	Value
Gateway	192.168.1.254
Subnet Mask	255.255.255.0
Primary DNS	inherited
Secondary DNS	inherited
Primary WINS	inherited
Secondary WINS	inherited
Primary NIS	inherited
Secondary NIS	inherited
Primary NTP	inherited
Secondary NTP	inherited
POP3 Server	inherited
SMTP Server	inherited
DNS Suffix	None

Custom DHCP options

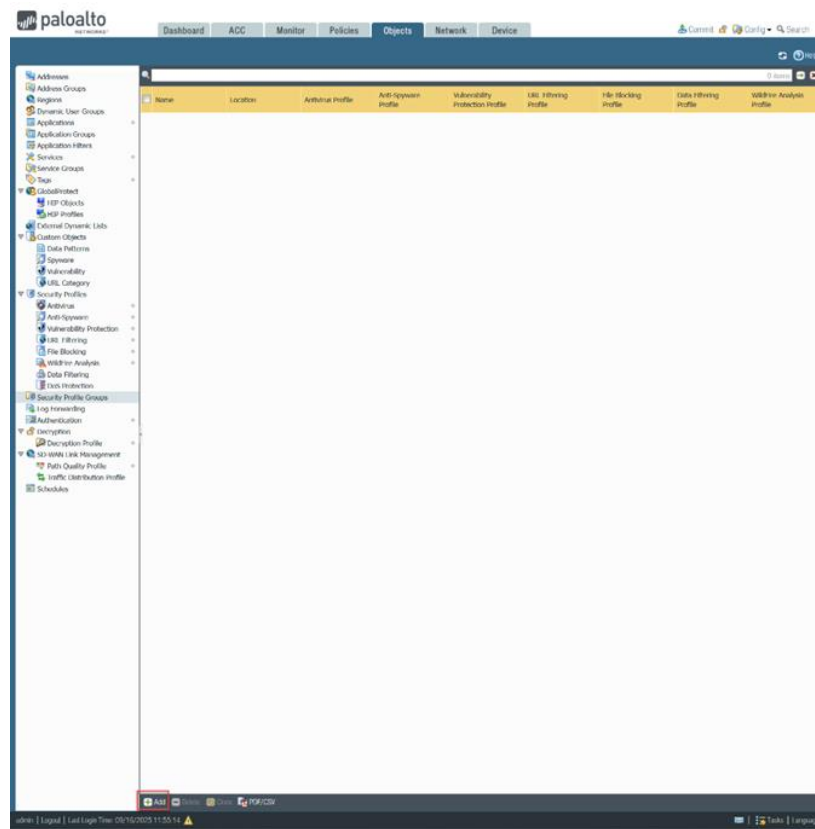
Name	Code	Type	Value
------	------	------	-------

Buttons: Add, Delete, Move Up, Move Down, OK, Cancel

5. Modify the settings as highlighted in the image above, except Change DNS Suffix to “inherited.” Then select the OK button.



5. Select the Security Profile Groups section on the left side of the Web GUI as highlighted in the image above.



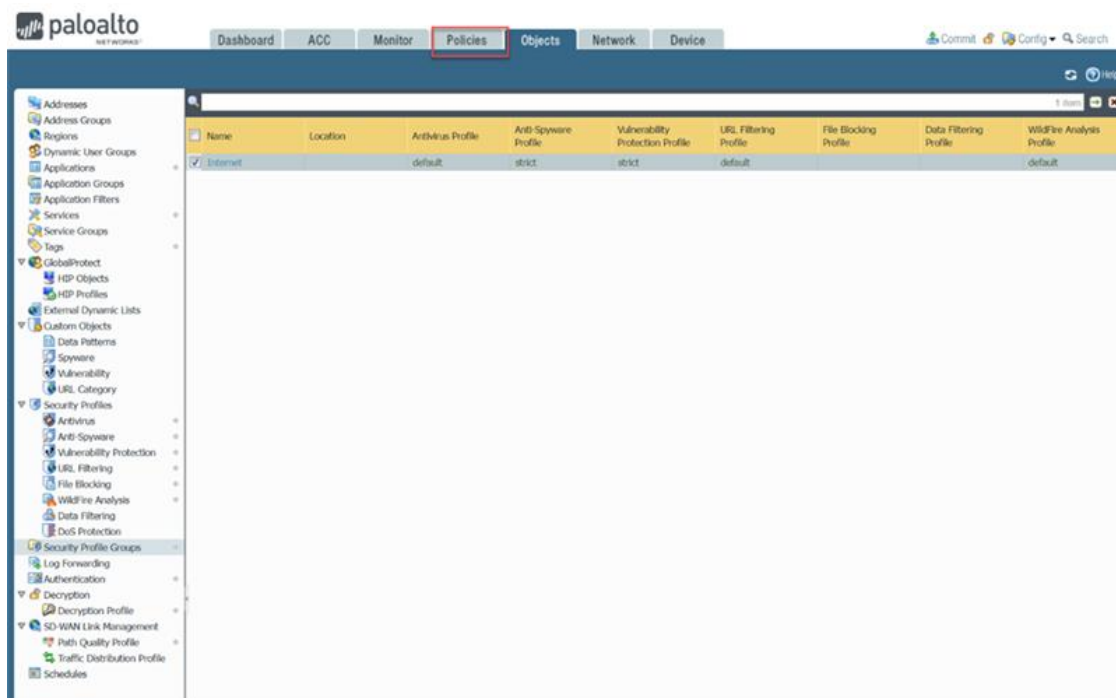
- Click the Add button on the bottom of the GUI as highlighted in the image above.

Security Profile Group

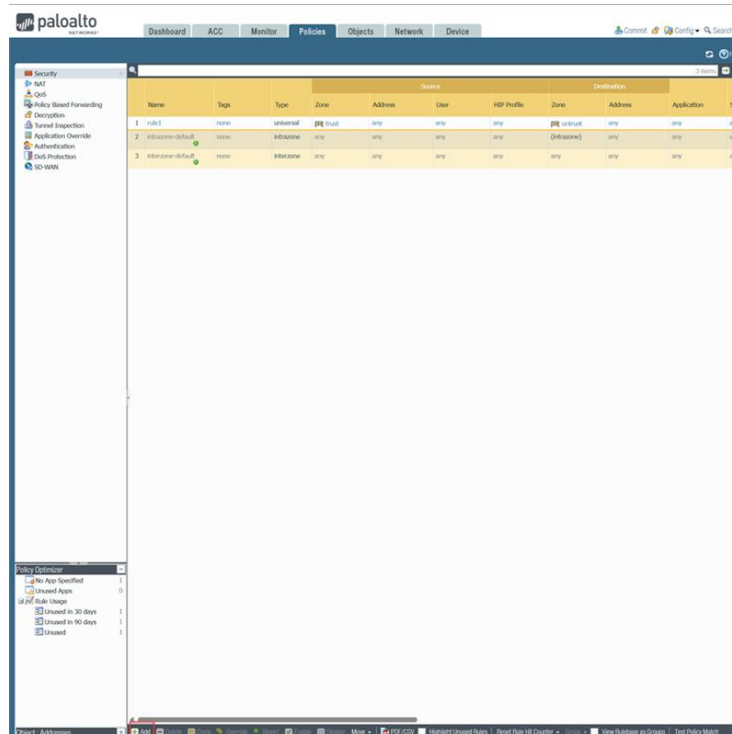
Name	Internet
Antivirus Profile	default
Anti-Spyware Profile	strict
Vulnerability Protection Profile	strict
URL Filtering Profile	default
File Blocking Profile	None
Data Filtering Profile	None
WildFire Analysis Profile	default

OK Cancel

- Modify the settings as highlighted in the image above. Then, select the OK button.



- Click the Policies tab on the top of the GUI as highlighted in the image above.



- Click the Add button as highlighted in the image above.

The screenshot shows the Security Policy Rule configuration window. The 'Source' tab is selected, and the 'Name' field is set to 'Internet Outgoing', 'Rule Type' is 'universal (default)', and 'Group Rules By Tag' is 'None'.

Security Policy Rule

General **Source** User Destination Application Service/URL Category Actions

Name: **Internet Outgoing**

Rule Type: **universal (default)**

Description:

Tags:

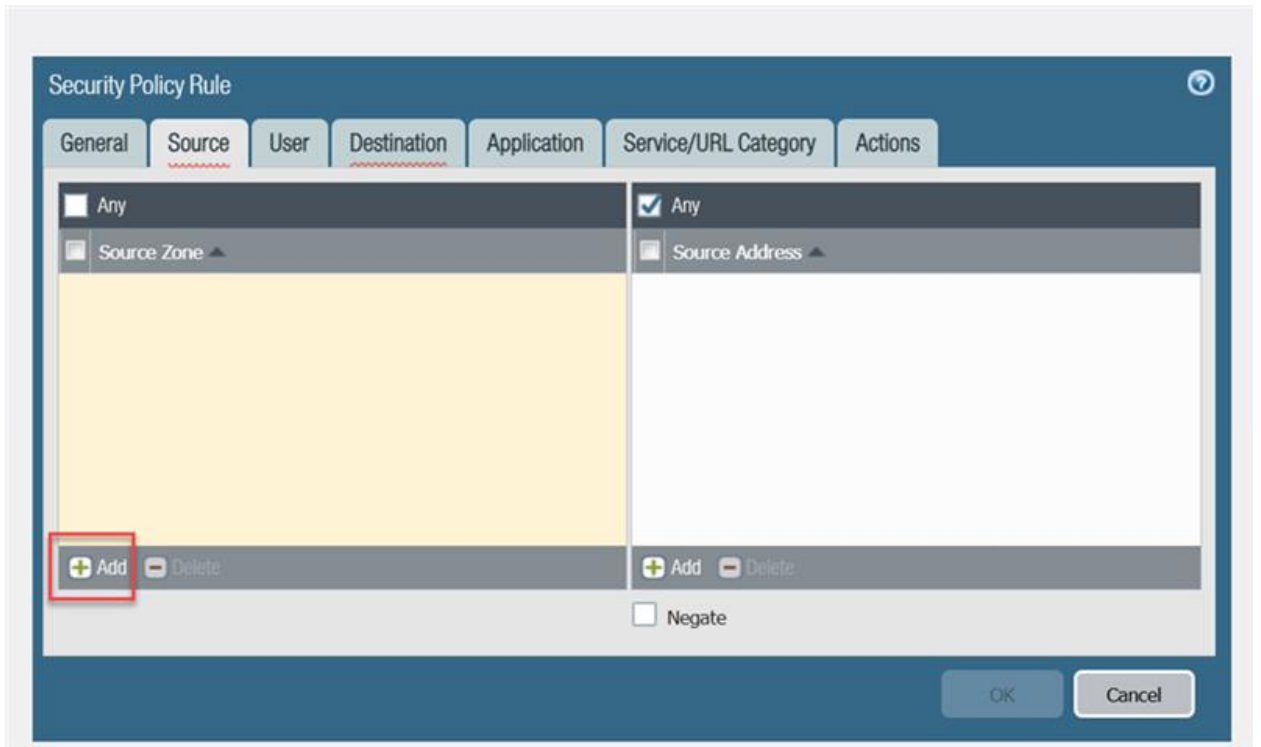
Group Rules By Tag: **None**

Audit Comment:

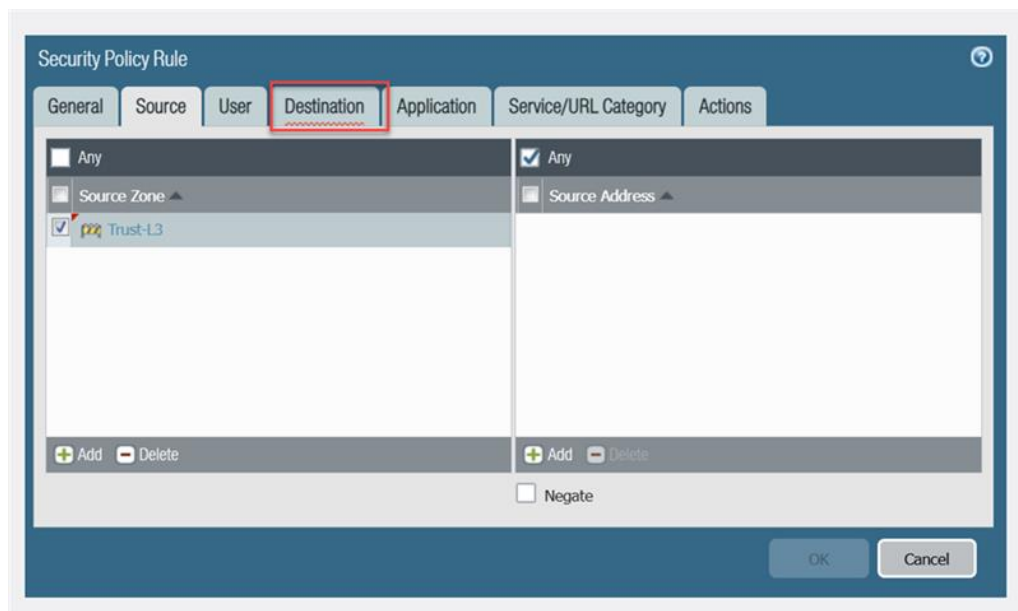
[Audit Comment Archive](#)

OK Cancel

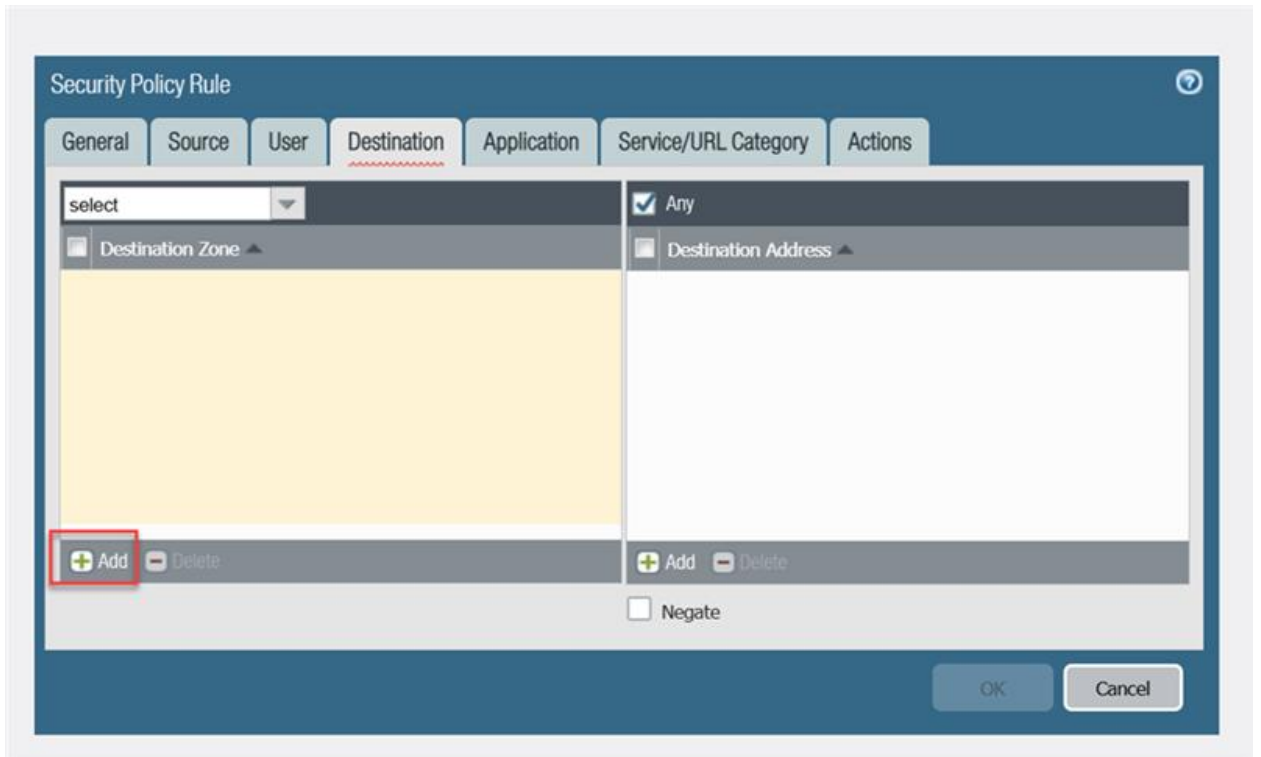
- Modify the settings as highlighted in the image above. Then, select the Source tab as highlighted in the image above.



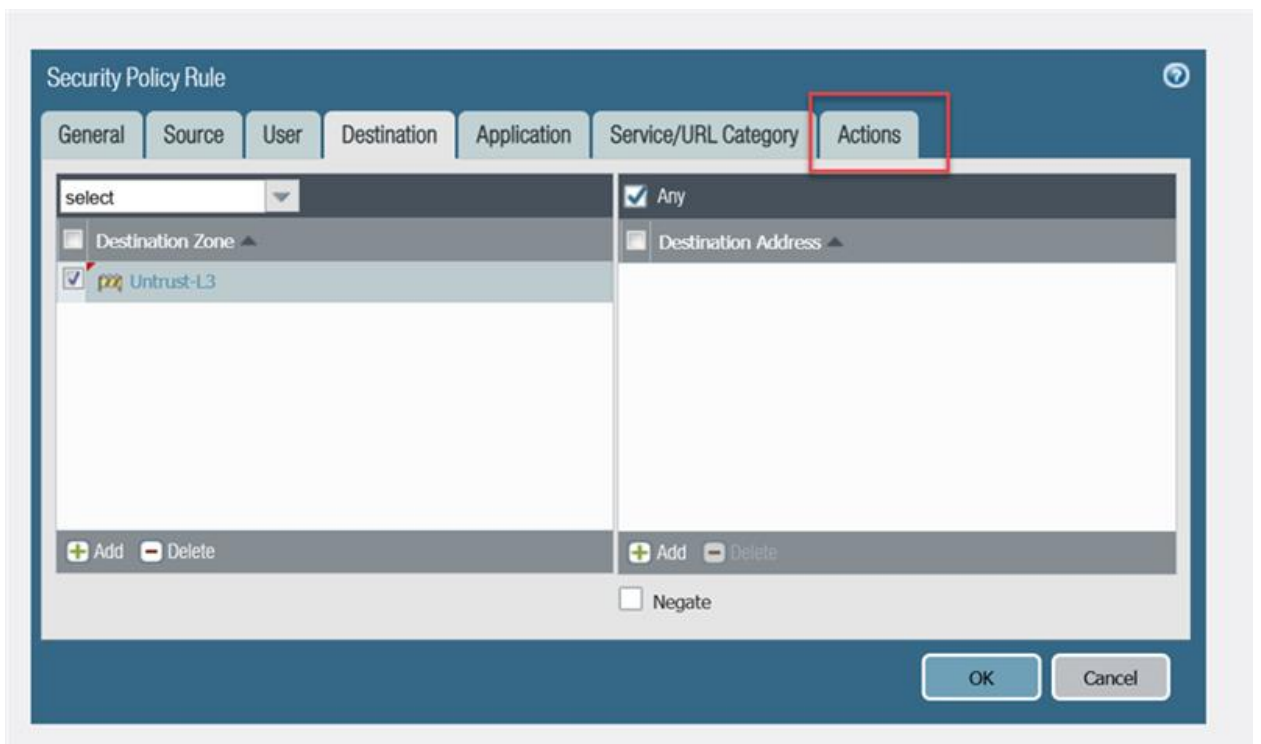
5. Select the Add button as highlighted in the image above. Then type in Trust-L3 before pressing the enter button.



5. Click on the Destination tab as highlighted in the image above.



5. Select the Add button as highlighted in the image above. Then type in Untrust-L3 before pressing the enter button.



5. Select the Actions tab as highlighted in the image above.

The screenshot shows the 'Security Policy Rule' configuration window with the 'Actions' tab selected. The 'Action' dropdown is set to 'Allow'. The 'Log at Session End' checkbox is checked. The 'Log Forwarding' dropdown is set to 'None'. The 'Profile Type' dropdown is set to 'Group' and the 'Group Profile' dropdown is set to 'Internet'. The 'Schedule' and 'QoS Marking' dropdowns are both set to 'None'. The 'OK' button is highlighted.

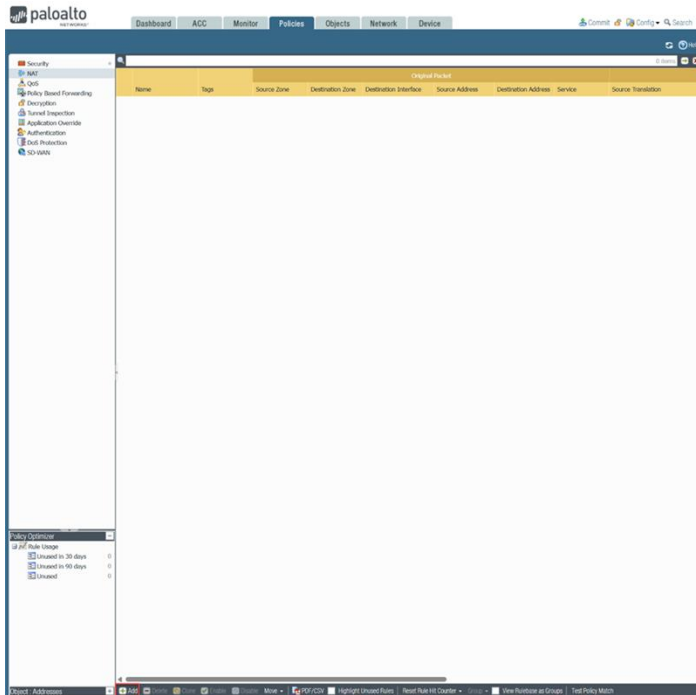
Section	Setting	Value
Action Setting	Action	Allow
	Send ICMP Unreachable	☐
Log Setting	Log at Session Start	☐
	Log at Session End	☒
Log Forwarding	Log Forwarding	None
Profile Setting	Profile Type	Group
	Group Profile	Internet
Other Settings	Schedule	None
	QoS Marking	None
Disable Server Response Inspection		☐

5. Modify the settings as highlighted in the image above. Then, select the OK button.

The screenshot shows the Palo Alto Networks GUI with the 'Policies' tab selected. The left sidebar shows the 'Security' section with 'NAT' highlighted. The main area displays a table of NAT rules.

Name		Tags	Type	Zone	Address	User	HIP Profile	Zone	Address	Application	Service
1	rule1	none	universal	trust	any	any	any	untrust	any	any	any
2	Internet Outgoing	none	universal	Trust-L3	any	any	any	Untrust-L3	any	any	any
3	Intrazone-default	none	Intrazone	any	any	any	any	(Intrazone)	any	any	any
4	Interzone-default	none	Interzone	any	any	any	any	any	any	any	any

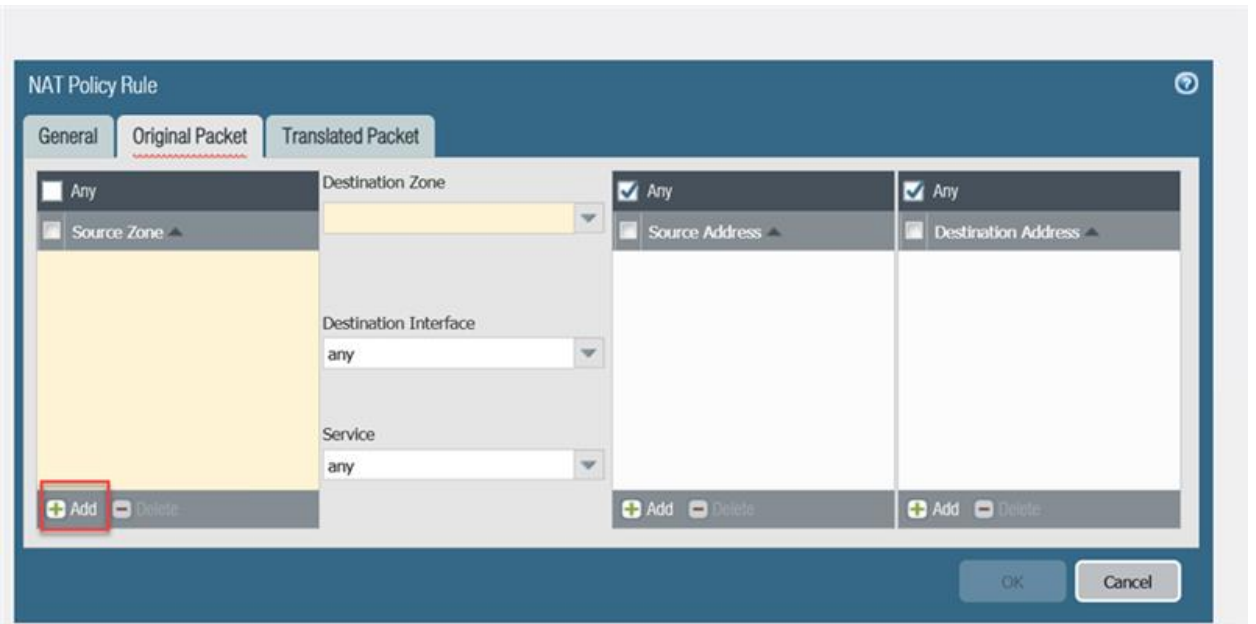
5. Select the NAT section on the left of the GUI as highlighted in the image above.



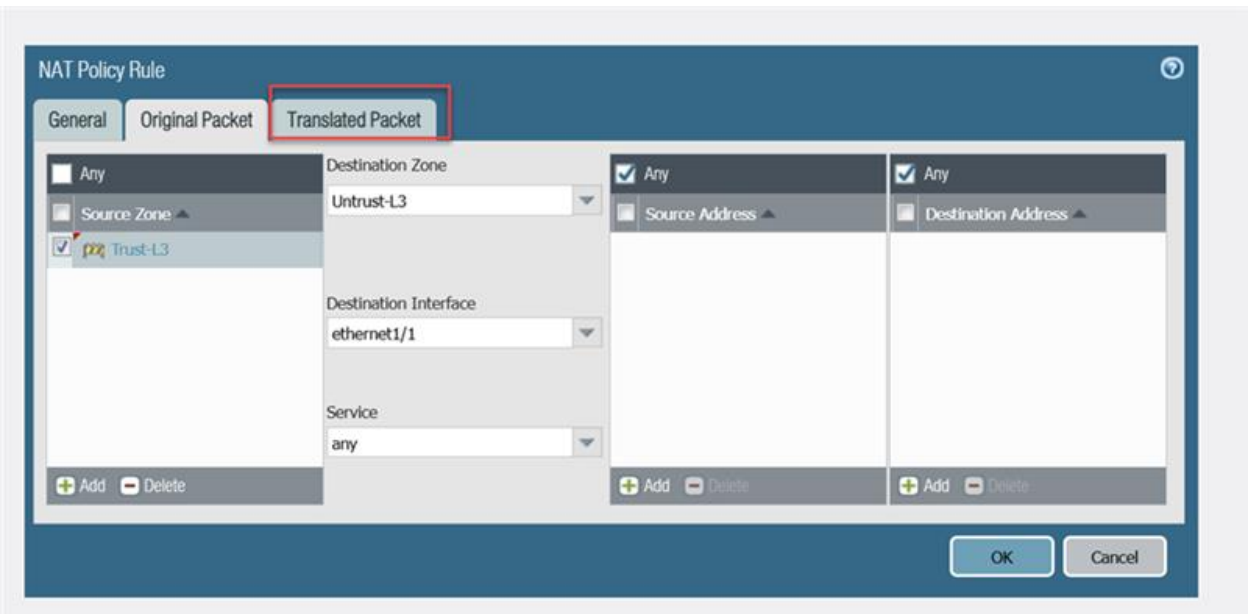
5. Select the Add button as highlighted in the image above.

The screenshot shows the NAT Policy Rule configuration dialog. The 'Original Packet' tab is selected. The 'Name' field is set to 'Internet Outgoing'. The 'NAT Type' is set to 'ipv4'. The 'Group Rules By Tag' is set to 'None'. The 'Audit Comment' field is empty. The 'OK' and 'Cancel' buttons are at the bottom right.

5. Modify the settings as highlighted in the image above. Then, select the Original Packet, as highlighted in the image above.



5. Click the Add button as highlighted in the image above. Type Trust-L3 before pressing the enter key.



5. Select the Translated Packet tab as highlighted in the image above.

The image shows the 'NAT Policy Rule' configuration window. The 'General' tab is selected. Under 'Source Address Translation', the 'Translation Type' is set to 'Dynamic IP And Port', 'Address Type' is 'Interface Address', 'Interface' is 'ethernet1/1', and 'IP Address' is 'None'. Under 'Destination Address Translation', the 'Translation Type' is set to 'None'. The 'OK' button is highlighted with a red box.

5. Modify the settings as highlighted in the image above. Then, select the OK button.

Dashboard

ACC

Monitor

Policies

Objects

Network

Device

Commit

Config

Search

Security

NAT

QoS

Policy Based Forwarding

Decryption

Tunnel Inspection

Application Override

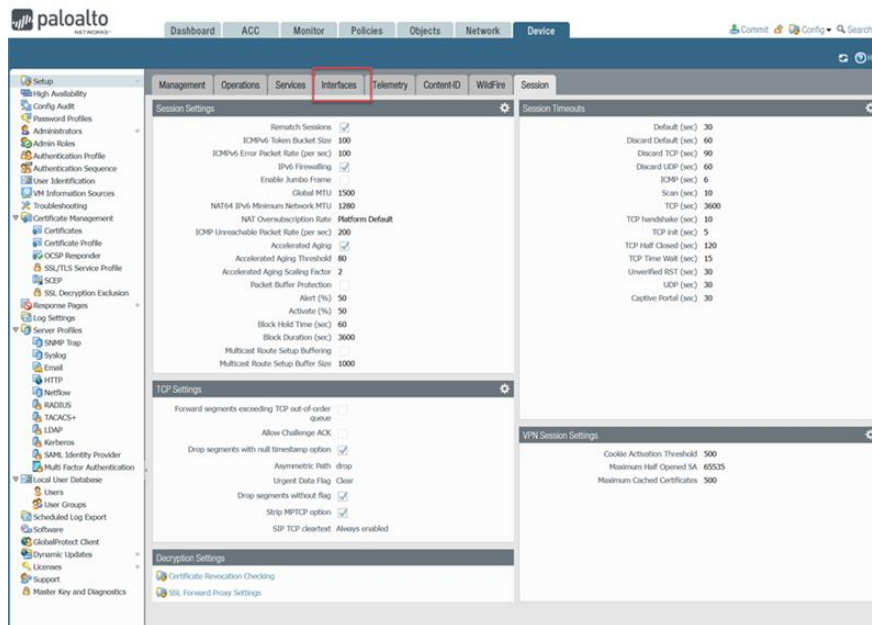
Authentication

DOS Protection

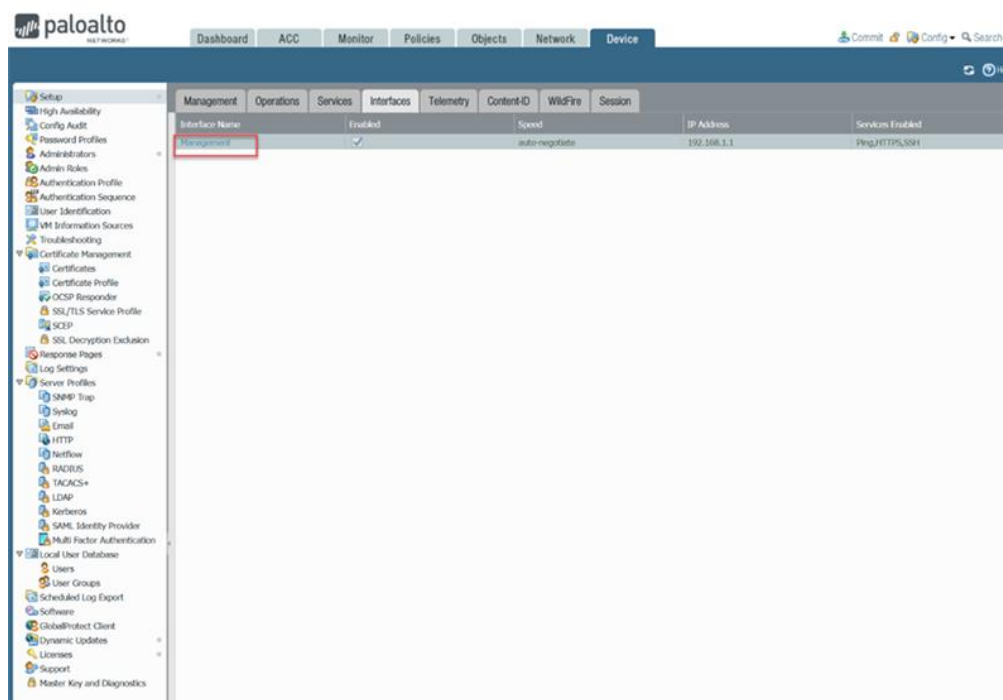
SD-WAN

		Original Packet							
Name	Tags	Source Zone	Destination Zone	Destination Interface	Source Address	Destination Address	Service	Source Translation	
1 Internet Outgoing	none	ppp Trust-L3	ppp Untrust-L3	ethernet1/1	any	any	any	dynamic ip-and-port ethernet1/1	

5. Select the Device tab on the top of the GUI as highlighted in the image above.



5. Check if all configurations in the Session tab are matching as in the image above. Then, select the Interfaces tab as highlighted in the image above.



5. Select the Management Interface as highlighted in the image above.

Management Interface Settings

IP Type: ☒ Static ☐ DHCP Client

IP Address: 192.168.1.1

Netmask: 255.255.255.0

Default Gateway: 192.168.1.254

IPv6 Address/Prefix Length:

Default IPv6 Gateway:

Speed: auto-negotiate

MTU: 1500

Administrative Management Services

☐ HTTP ☒ HTTPS

☐ Telnet ☒ SSH

Network Services

☐ HTTP OCSP ☒ Ping

☐ SNMP ☐ User-ID

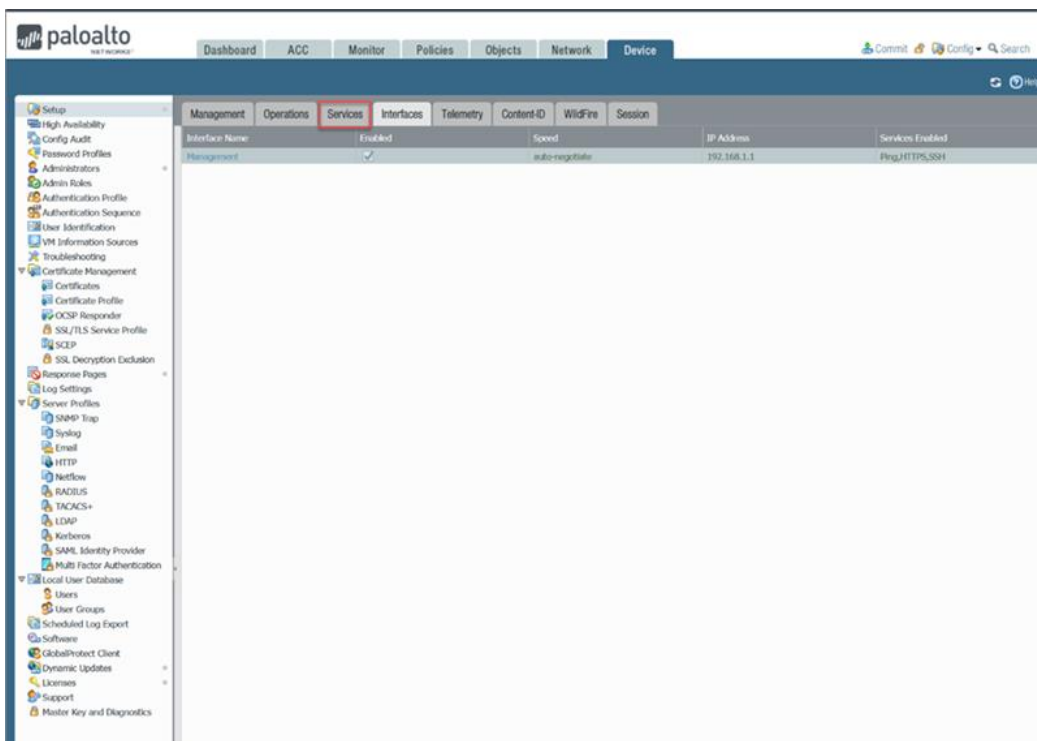
☐ User-ID Syslog Listener-SSL ☐ User-ID Syslog Listener-UDP

Permitted IP Addresses:

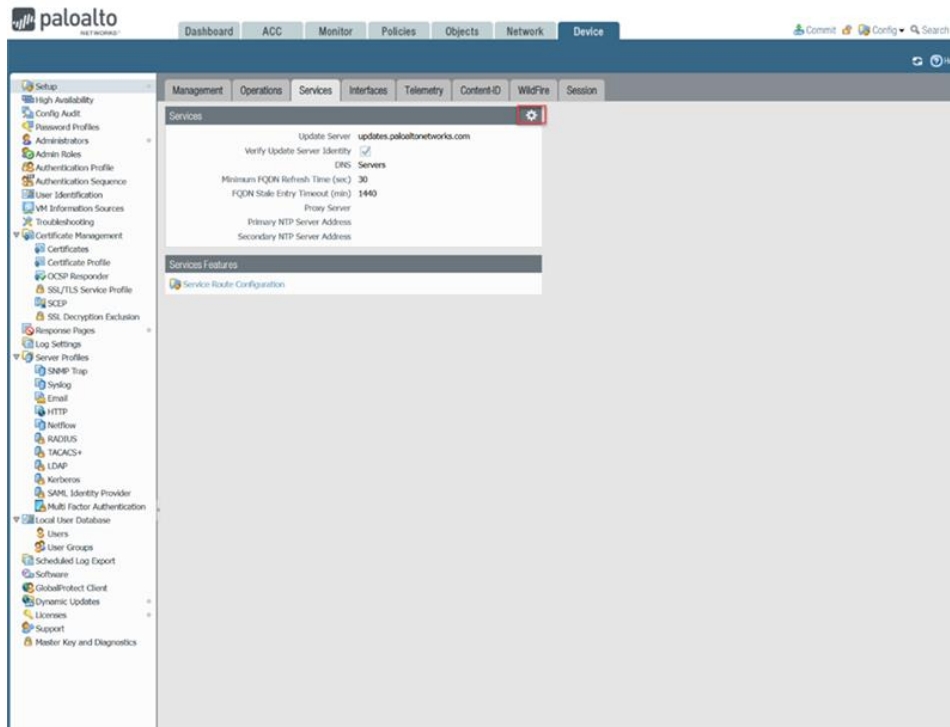
Description:

Buttons: Add, Delete, OK, Cancel

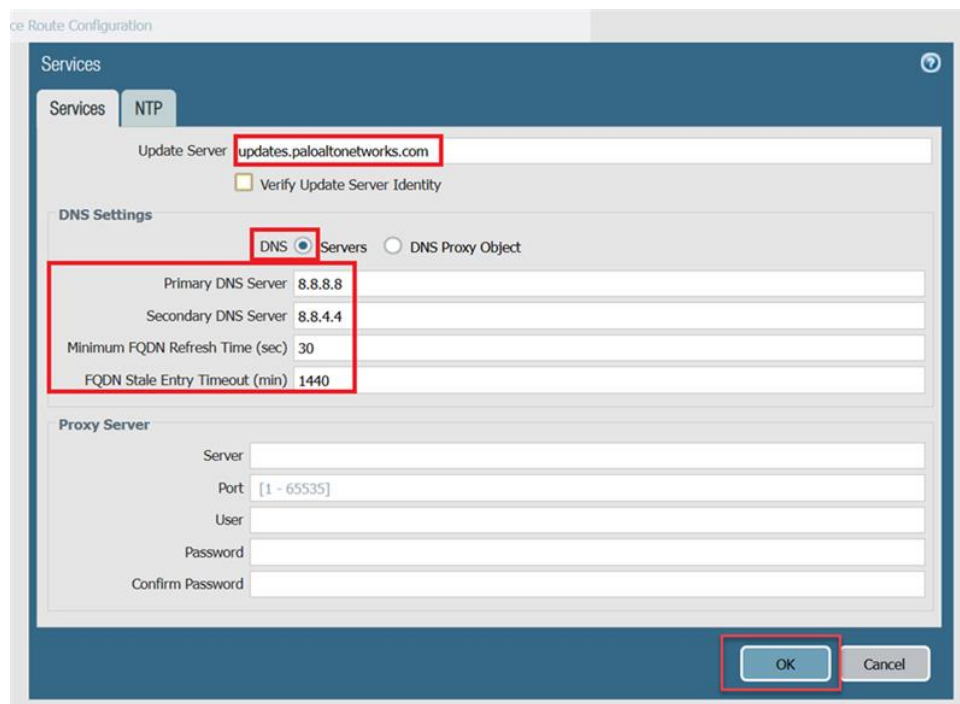
5. Modify the settings as highlighted in the image above. Then, select the OK button as highlighted in the image above.



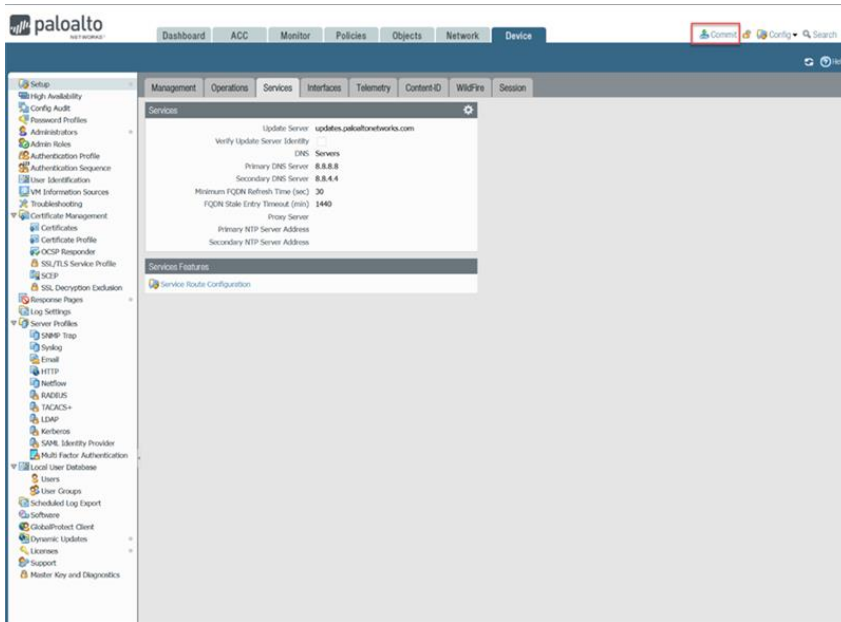
5. Select the Services tab as highlighted in the image above.



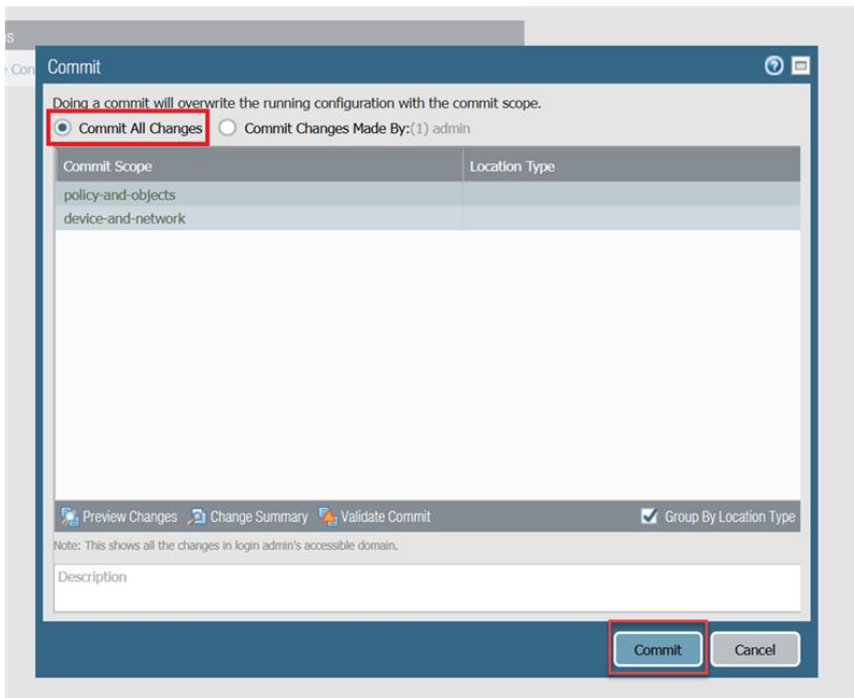
- Click on the System icon as highlighted in the image above.



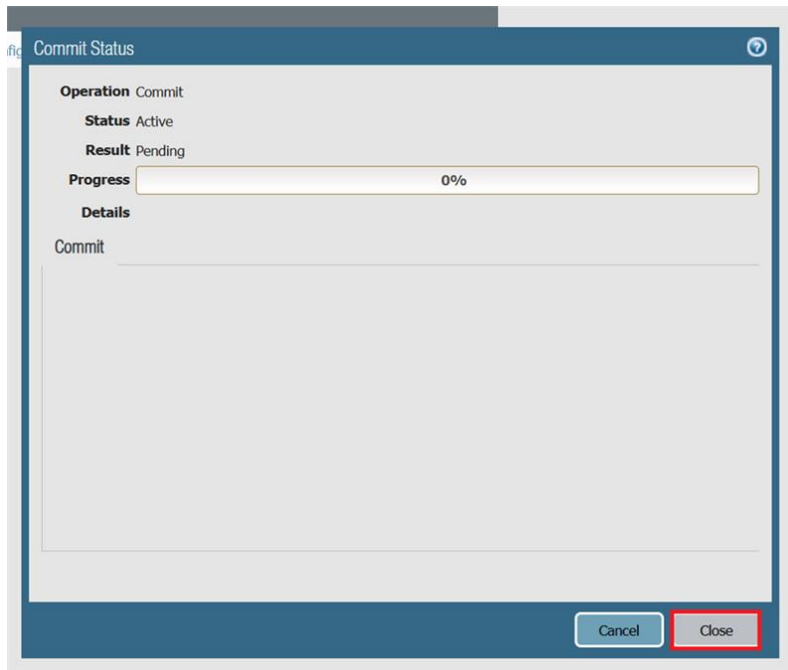
- Modify the settings as highlighted in the image above. Then, select the OK button.



5. Press the Commit button as highlighted in the image above in the top right corner of the GUI.



5. Select the Commit All Changes option as highlighted in the image above. Then, select the Commit button.



5. When the Progress of the Commit reaches 100% press the Close button as highlighted in the image above

Problems

We had to update Pan-OS from 9.1 to 10.2.9, so we had to install and download each version, which took 30 minutes. When we got the new version, we lost the device certificate, so we had to generate a one-time password and obtain a certificate from Palo-Alto which delayed the completion of this lab. We also had to re-configure the SOHO on the new OS.

Our WAN port leading to internet switch had layer-1 problems relating to the ethernet cable.

Conclusion

The PA220's web interface was confusing to navigate at first, but after configuring the firewall for a SOHO network, I believe I have gained a strong understanding of how to navigate PAN-OS's interface. All in all, a SOHO network configuration using a PA220 firewall had a relatively simple setup process and would work great for an individual or small business